

Console

Gate, compressor, limiter and output transformer from the centre section of an 80s British broadcast console.



Overview

Console recreates the dynamics section of an 80s British broadcast desk, in the spirit of the classic desk VCA compressor and its companion compressor / gate module. It is fast, clean and transparent: it holds the level without dulling the transients, and the console's output transformer adds a little weight.

The signal runs through four parts in order: the gate / expander, the compressor, a fast peak limiter and the output transformer. MIX blends the dry signal back in for parallel compression.

Reach for it for

- Tight, controlled drums - with the gate cleaning up spill
- Spoken voice and presenters that need to stay steady and present
- Bass DI and acoustic instruments that want even level without losing attack
- Parallel squash on a drum or mix bus
- A touch of transformer weight on its own (Transformer Only)

Controls

Gate / Expander

THRESH	The level below which the gate or expander starts turning the signal down, -80 to 0 dB. Fully left is OFF.
RANGE	How far it turns the signal down, 0 - 60 dB. Small settings just lower the background; 40 dB or more shuts it right off.
RELEASE	How fast it closes once the signal falls below the threshold, 20 - 2000 ms. It opens very fast (in about half a millisecond), so the attack of a hit is kept.
IN	Switches the gate / expander in. Off by default.
GATE	Shuts by the full RANGE as soon as the signal falls below the threshold. Tight and obvious - for drums and spill.
EXP	A gentle 1:2 expander: it only takes down what falls below the threshold, and the further below, the more it takes down (up to RANGE). Smoother and more natural - for noise, room and breaths. This is the default.
OPEN	Lights while the gate is letting the signal through.
GATE readout	How far the gate is turning the signal down right now, in dB (GATE OUT when the section is off).

Compressor

A fast VCA compressor with a 3 dB soft knee and a quick RMS detector. Its detector hears the gated signal, so a closed gate does not make it work.

THRESH	The level where the compressor starts working, -40 to 0 dB. Turn it down for more compression.
RATIO	How strongly it compresses once over the threshold, 1:1 to INF:1 (at the top of the knob: limiting).
ATTACK	How fast it clamps down, 0.1 - 30 ms. Short settings catch the transients; longer ones let them through for more punch.

RELEASE	How fast it lets go, 30 - 3000 ms.
GAIN	Make-up gain, -10 to +20 dB. Only applies while the compressor is in.
IN	Switches the compressor in or out.
GR meter	The row of 12 lights shows the gain reduction of the compressor and limiter together, from 1 to 30 dB.

Limit / Output

LIMIT	A fast peak limiter after the compressor, -20 to 0 dB: peaks above this level are held at it. Fully right is OFF. The red light beside the knob shows it limiting.
XFMR	The console's output transformer. On (default): the lows thicken and saturate gently as the level rises, and the extreme top softens. Off: a straight wire.
LINK	On (default): both sides share one gain in the gate, compressor and limiter, so the stereo image stays still. Off: the two sides work as two mono channels.
MIX	Parallel compression: blends the dry signal back in. 100% (WET 100) is fully processed; lower settings keep the dry transients under the processed sound.
IN / OUT meters	Input and output level.
GR readout	The total gain reduction of the compressor and limiter, in dB.

The header

PRESETS	Opens the list of factory presets; the ◀ / ▶ keys step through them and the display shows the one in use. Choosing one sets the gate, compressor, limiter, XFMR, LINK and MIX. The last entry resets everything to the defaults.
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S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Input guard

Console has no input or output fold-out: its own transformer and limiter do that job. An input guard runs in front of everything and is on by default: it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. It has no key on the panel; it appears as Input guard in REAPER's parameter list.

Presets

Desk Glue	Gentle 2.5:1 compression with a little make-up gain; gate off.
Presenter Voice	A soft expander under a firm 4:1 compressor with fast attack and the limiter at -2 dB - steady, upfront speech.
Tight Drums (gated)	A hard gate with 40 dB range and a fast 5:1 compressor, limiter at -1 dB.
Kick / Snare Gate	A deep, fast gate (60 dB range, 80 ms release), 4:1 compression and LINK off for two independent channels.
Parallel Squash	Heavy 8:1 compression with very fast timing and the limiter at -3 dB, blended in at 45%.
Transmitter Limit	INF:1 with the fastest attack and release - a broadcast-style brickwall at -14 dB, limiter at -1 dB.
Acoustic Clean-up	A gentle expander with a slow release under 3:1 compression with a 10 ms attack.
Bass DI	4:1 compression with an 8 ms attack to keep the pluck; gate off.

Transformer Only	Gate and compressor out: just the output transformer's colour.
Gentle Programme Bus	A light expander and slow, gentle 2:1 compression with the limiter at -1 dB.
Reset all to defaults	Every control back to its starting value.

Tips

- Set the gate first with the compressor out, then bring the compressor in: the compressor listens after the gate, so the two settings work together.
- If the gate chatters on decaying notes, try EXP instead of GATE, or lengthen the RELEASE.
- For more punch, raise the compressor's ATTACK to 10 - 30 ms so the front of each hit gets through.
- Use LIMIT as a safety net a few dB below your peaks after a heavy compressor setting.
- On a stereo bus keep LINK on; switch it off only for dual-mono sources such as two separate drum mics.

File: gfx-console.jsfx